AI Medical Receptionist

Project Brief

Problem Statement

Hospital reception teams handle many routine calls about doctor availability, appointment booking, confirmations, and cancellations. Managing these requests manually can take staff time and may cause delays for patients trying to arrange an appointment. An AI-powered voice receptionist can provide a convenient first point of contact, collect required appointment details, and use backend tools to check availability and process appointment requests.

Features

- Voice-based interaction through a web frontend.
- Collect the patient's full name, preferred appointment date, time, and reason for visit.
- Check doctor availability before booking.
- Book appointments after validating details and receiving patient confirmation.
- Reject past dates and appointment times that have already passed.
- Confirm existing appointment details when supported by the backend.
- Process appointment cancellation requests.
- Display call status and conversation transcription in the frontend.
- Use configurable speech-to-text, language-model, and text-to-speech providers through Vapi.
- Support multiple languages when the selected models and voices support them.

Suggested Technology Stack

Voice orchestration: Vapi **Backend:** FastAPI **Database:** SQLite or PostgreSQL, depending on the deployed configuration **Frontend:** Web interface

Safety and privacy note: Use test data during development. Before handling real patient information, confirm applicable privacy requirements, provider agreements, access controls, retention settings, and whether the complete deployment meets the hospital's security and compliance requirements. AI responses should not replace clinical advice.